
Polynomials: Basics

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CONTENTS

1	Roots	2
2	Quadratic Formula	2
3	Vieta’s Formulas	2
4	Factorizing	3
5	Selected Problems	4
6	Solutions	5
6.1	2024 HMMT	5
6.2	1998 HMMT	6
6.3	2009 MPfG	7
6.4	2025 FLSAM	8
6.5	2020 HMMT	9
6.6	2024 CMIMC	10
6.7	2006 HMMT	11
6.8	2025 FLSAM	12

1 ROOTS

Consider the polynomial

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0.$$

Then, this polynomial has n roots and a degree of n . These roots can be real or complex. Say $r_1, r_2, \dots, r_n$ are roots. Then, $P(r_i) = 0$ for $1 \leq i \leq n$. Furthermore, for each r_i with $1 \leq i \leq n$, there exists a polynomial $Q(x)$ with degree $n - 1$ such that

$$P(x) = (x - r_i) \cdot Q(x).$$

2 QUADRATIC FORMULA

Let $P(x) = ax^2 + bx + c$ be a quadratic polynomial. Then, we will use a technique known as *completing the square* to find the roots. Notice, multiplying by $4a$, then:

$$\begin{aligned} ax^2 + bx + c &= 0 \\ 4a^2 x^2 + 4abx + 4ac &= 0 \end{aligned}$$

Now, we will perform the *complete the square*:

$$\begin{aligned} 4a^2 x^2 + 4abx + 4ac &= 0 \\ (2ax + b)^2 + (4ac - b^2) &= 0 \end{aligned}$$

From here, we can solve for x :

$$(2ax + b)^2 = b^2 - 4ac \implies 2ax + b = \pm \sqrt{b^2 - 4ac} \implies x = \boxed{\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}.$$

This is exactly the quadratic formula. There are a few important corollaries. Firstly, when $b^2 - 4ac > 0$, then two real roots exist. When $b^2 - 4ac = 0$, then only one root exists. But, if $b^2 - 4ac < 0$, then no real roots exist. In fact, the roots will be complex.

3 VIETA'S FORMULAS

Consider the polynomial

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

with complex coefficients and roots $r_1, r_2, \dots, r_n$. Define s_j to be the j^{th} elementary symmetric polynomial of the roots. Then:

$$\begin{aligned} s_1 &= r_1 + r_2 + \cdots + r_n = (-1)^1 \frac{a_{n-1}}{a_n}, \\ s_2 &= r_1 r_2 + r_2 r_3 + \cdots + r_{n-1} r_n = (-1)^2 \frac{a_{n-2}}{a_n}, \\ &\vdots \\ s_n &= r_1 r_2 r_3 \cdots r_n = (-1)^n \frac{a_0}{a_n}. \end{aligned}$$

We will look at this for $n = 2$. Suppose $P(x)$ has roots r_1 and r_2 . Then:

$$\begin{aligned} P(x) &= C \cdot (x - r_1)(x - r_2) \\ &= C \cdot (x^2 - (r_1 + r_2)x + r_1 r_2) \\ &= C \cdot x^2 + C \cdot (-(r_1 + r_2)) \cdot x + C \cdot (r_1 r_2) \end{aligned}$$

Thus, if $P(x) = a_2x^2 + a_1x + a_0$, then:

$$a_2 = C$$

$$a_1 = C \cdot -(r_1 + r_2)$$

$$a_0 = C \cdot (r_1 r_2)$$

Thus, $r_1 + r_2 = (-1)^1 \frac{a_1}{a_2}$ and $r_1 r_2 = (-1)^2 \frac{a_0}{a_2}$. This is identical to the Vieta's Formulas. One can see this is true for $n > 2$, but crucially, when expanding the $P(x) = (x - r_1)(x - r_2) \cdots (x - r_n)$, then the coefficients of the polynomial are the n symmetric sums.

4 FACTORIZING

Usually, the goal is to factor things into *symmetric polynomials*. These are a few of some common tricks:

- $(a + b)^2 = a^2 + 2ab + b^2$
- $(a + b)(a - b) = a^2 - b^2$.
- $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
- $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$
- $(a + b + c)^3 = a^3 + b^3 + c^3 + 3(a + b)(b + c)(c + a)$

5 SELECTED PROBLEMS

Problem 1. (2024 HMMT) Suppose r , s , and t are nonzero reals such that the polynomial $x^2 + rx + s$ has s and t as roots, and the polynomial $x^2 + tx + r$ has 5 as a root. Compute s .

Problem 2. (1998 HMMT) Given that three roots of $f(x) = x^4 + ax^2 + bx + c$ are 2, -3 , and 5, what is the value of $a + b + c$?

Problem 3. (2009 MPfG) The three roots of the cubic $30x^3 - 50x^2 + 22x - 1$ are distinct real numbers between 0 and 1. For every nonnegative integer n , let s_n be the sum of the n th powers of these three roots. What is the value of the infinite series

$$s_0 + s_1 + s_2 + s_3 + \cdots ?$$

Problem 4. (2025 FLSAM) Distinct reals a, b, c are the roots of the cubic $x^3 - 10x^2 + kx - 2$, for some real k . Given that $ax^2 + bx + c = 0$ has a unique real solution, compute k .

Problem 5. (2020 HMMT) Let $P(x) = x^3 + x^2 - r^2x - 2020$ be a polynomial with roots r, s, t . What is $P(1)$?

Problem 6. (2024 CMIMC) Suppose $P(x) = x^2 + Ax + B$ for real A and B . If the sum of the roots of $P(2x)$ is $\frac{1}{2}$ and the product of the roots of $P(3x)$ is $\frac{1}{3}$, find $A + B$.

Problem 7. (2006 HMMT) Let a, b, c be the roots of $x^3 - 9x^2 + 11x - 1 = 0$, and let $s = \sqrt{a} + \sqrt{b} + \sqrt{c}$. Find $s^4 - 18s^2 - 8s$.

Problem 8. (2025 FLSAM) The graphs of $y = x^3 - 5x^2 + 7x + 4$ and $y = 2x^2 - 6$ intersect at A , B , and C . Find the sum of the coordinates of the centroid of triangle ABC .

6 SOLUTIONS

6.1 2024 HMMT

Suppose r , s , and t are nonzero reals such that the polynomial $x^2 + rx + s$ has s and t as roots, and the polynomial $x^2 + tx + r$ has 5 as a root. Compute s .

Solution. By Vieta on the first quadratic, we have

$$st = s \implies t = 1,$$

because we are told that r, s, t are nonzero.

Then, let q be the second root in the second quadratic. By Vieta, we have

$$\begin{aligned} 5 + q = -t &\implies 5 + q = -1 \\ &\implies q = -6. \end{aligned}$$

Now, applying Vieta again but on the product of the roots in the second quadratic, we get

$$s = 5 \cdot (-6) = -30.$$

Finally, applying Vieta on the sum of the roots of the first quadratic gives

$$\begin{aligned} s + t = -r &\implies r = -s - t \\ &\implies r = 30 - 1 = \boxed{29}. \end{aligned}$$

6.2 1998 HMMT

Given that three roots of $f(x) = x^4 + ax^2 + bx + c$ are 2, -3 , and 5, what is the value of $a + b + c$?

Solution. We see that the coefficient of the x^3 term of the polynomial is 0, so the sum of the roots of $f(x)$ is 0. This means that the fourth root is -4 . We can now write $f(x) = (x - 2)(x + 3)(x + 4)(x - 5)$. Plugging in $x = 1$ will give us the sum of the coefficients of $f(x)$, so we have

$$f(1) = (1 - 2)(1 + 3)(1 + 4)(1 - 5) = (-1)(4)(5)(-4) = 80.$$

Subtracting the coefficient of the x^4 term gives us that $a + b + c = 80 - 1 = \boxed{79}$.

6.3 2009 MPFG

The three roots of the cubic $30x^3 - 50x^2 + 22x - 1$ are distinct real numbers between 0 and 1. For every nonnegative integer n , let s_n be the sum of the n th powers of these three roots. What is the value of the infinite series

$$s_0 + s_1 + s_2 + s_3 + \dots?$$

Solution. Let the three roots be a, b, c . Then we want

$$(1 + 1 + 1) + (a + b + c) + (a^2 + b^2 + c^2) + (a^3 + b^3 + c^3) + \dots.$$

This is a geometric sequence which is equivalent to:

$$\frac{a}{1-a} + \frac{b}{1-b} + \frac{c}{1-c}.$$

We can expand this:

$$\begin{aligned} \frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} &= \frac{(1-a)(1-b) + (1-b)(1-c) + (1-c)(1-a)}{(1-a)(1-b)(1-c)} \\ &= \frac{3 - 2(a+b+c) + (ab+bc+ca)}{(1-a)(1-b)(1-c)}. \end{aligned}$$

By Vieta's, $a + b + c = \frac{50}{30}$ and $ab + bc + ca = \frac{22}{30}$. Also,

$$(1-a)(1-b)(1-c) = \frac{P(1)}{30} = \frac{30 - 50 + 22 - 1}{30} = \frac{1}{30}.$$

Therefore:

$$\frac{3 - 2(a+b+c) + (ab+bc+ca)}{(1-a)(1-b)(1-c)} = \frac{3 - 2 \cdot (50/30) + (22/30)}{(1/30)} = \frac{90 - 2 \cdot 50 + 22}{1} = \boxed{12}.$$

6.4 2025 FLSAM

Distinct reals a, b, c are the roots of the cubic $x^3 - 10x^2 + kx - 2$, for some real k . Given that $ax^2 + bx + c = 0$ has a unique real solution, compute k .

Solution. Taking the discriminant, we must have $b^2 = 4ac$. Now by Vieta formulas, we have

$$\begin{aligned}a + b + c &= 10, \\ab + ac + bc &= k, \\abc &= 2.\end{aligned}$$

Then $b^3 = 4abc = 8$ and so $b = 2$. We now have

$$k = ab + ac + bc = b(a + c) + \frac{abc}{b} = 2(10 - 2) + \frac{2}{2} = \boxed{17}$$

6.5 2020 HMMT

Let $P(x) = x^3 + x^2 - r^2x - 2020$ be a polynomial with roots r, s, t . What is $P(1)$?

Solution. Since r is a root of $P(x)$, we get

$$\begin{aligned}P(r) = 0 &\implies r^3 + r^2 - r^3 - 2020 = 0 \\&\implies r^2 = 2020.\end{aligned}$$

Therefore, the answer is

$$\begin{aligned}P(1) &= 1 + 1 - r^2 - 2020 \\&= 1 + 1 - 2020 - 2020 \\&= \boxed{4038}.\end{aligned}$$

6.6 2024 CMIMC

Suppose $P(x) = x^2 + Ax + B$ for real A and B . If the sum of the roots of $P(2x)$ is $\frac{1}{2}$ and the product of the roots of $P(3x)$ is $\frac{1}{3}$, find $A + B$.

Solution. Note that

$$P(2x) = 4x^2 + 2A + B,$$

and Vieta tells us that the sum of the roots is $-\frac{A}{2}$. Hence, we have

$$-\frac{A}{2} = \frac{1}{2} \implies A = -1.$$

Also, note that

$$P(3x) = 9x^2 + 3A + B,$$

and Vieta tells us that the product of the roots is $\frac{B}{9}$. Therefore, we have

$$\frac{B}{9} = \frac{1}{3} \implies B = 3.$$

Hence, the answer is $A + B = -1 + 3 = \boxed{2}$.

6.7 2006 HMMT

Let a, b, c be the roots of $x^3 - 9x^2 + 11x - 1 = 0$, and let $s = \sqrt{a} + \sqrt{b} + \sqrt{c}$. Find $s^4 - 18s^2 - 8s$.

Solution. Our most natural instinct is to square the equation $s = \sqrt{a} + \sqrt{b} + \sqrt{c}$, which gives us

$$s^2 = (\sqrt{a} + \sqrt{b} + \sqrt{c})^2 = a + b + c + 2\sqrt{ab} + 2\sqrt{bc} + 2\sqrt{ca}.$$

We have a term of $a + b + c = 9$ on the right-hand side (which follows from Vieta's Formulas), and the $s^4 - 18s^2$ within the desired expression motivates to bring this 9 onto the left-hand side and square again. This yields

$$\begin{aligned} (s^2 - 9)^2 &= (2\sqrt{ab} + 2\sqrt{bc} + 2\sqrt{ca})^2 \\ s^4 - 18s^2 + 81 &= 4ab + 4bc + 4ca + 8\sqrt{a^2bc} + 8\sqrt{ab^2c} + 8\sqrt{abc^2}. \end{aligned}$$

Consider the term of the last three terms on the right-hand side. Factoring out $8\sqrt{abc}$, we have

$$8\sqrt{abc}(\sqrt{a} + \sqrt{b} + \sqrt{c}) = 8\sqrt{1}s = 8s,$$

the $abc = 1$ from another application of Vieta's Formulas, and $ab + bc + ca = 11 \leftrightarrow 4ab + 4bc + 4ca = 44$, again from Vieta's, so

$$s^4 - 18s^2 + 81 = 44 + 8s \leftrightarrow s^4 - 18s^2 - 8s = \boxed{-37},$$

the requested answer.

6.8 2025 FLSAM

The graphs of $y = x^3 - 5x^2 + 7x + 4$ and $y = 2x^2 - 6$ intersect at A , B , and C . Find the sum of the coordinates of the centroid of triangle ABC .

Solution. Let (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) be the points of intersection. Then, x_1 , x_2 , and x_3 must satisfy

$$x^3 - 5x^2 + 7x + 4 = 2x^2 - 6 \iff x^3 - 7x^2 + 7x + 10 = 0.$$

Thus, by Vieta's, $x_1 + x_2 + x_3 = 7$ and $x_1x_2 + x_2x_3 + x_3x_1 = 7$. Therefore,

$$\begin{aligned} \sum_{\text{cyc}} y_i &= \sum_{\text{cyc}} (2x_i^2 - 6) \\ &= 2 \cdot \sum_{\text{cyc}} x_i^2 - 18 \\ &= 2 \cdot ((x_1 + x_2 + x_3)^2 - 2 \cdot (x_1x_2 + x_2x_3 + x_3x_1)) - 18 \\ &= 52. \end{aligned}$$

Hence, the sum of the coordinates of the centroid is equivalent to

$$\frac{1}{3} \cdot \left(\sum_{\text{cyc}} x_i + \sum_{\text{cyc}} y_i \right) = \boxed{\frac{59}{3}}$$